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Design and simulation of a broadband tunable transmittance optical filter using Sb₂Se₃ phase change material

SURAWUT WICHARN,¹ CHETTANAT PADUNGATTHAKIJ,² SUEBTARKUL SUCHAT,³ NATHAN HAGEN,⁴ AND PRATHAN BURANASIRI^{2,*}

¹ Department of Physics, Faculty of Science, Srinakharinwirot University, Bangkok, Thailand

² Department of Physics, School of Science, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand

³ Department of Physics, Faculty of Science and Technology, Phranakorn Rajabhat University, Bangkok, Thailand.

⁴ Center of Optical Research and Education & School of Engineering, Utsunomiya University, Utsunomiya, Japan.

*prathan.bu@kmitl.ac.th

Abstract: We propose a switchable optical filter design to be operated around 1064-nm wavelengths using antimony triselenide (Sb₂Se₃). Sb₂Se₃ is a phase change material (PCM) which has low absorption in infrared (IR) spectrum, high and stable refractive index contrast between the two phases from visible through IR spectrum, and a high optical endurance in an order of 10⁶ cycles. We give the reasoning for selecting long-pass edge filter design and compares its performance to other common planar thin-film designs. At 1064-nm, the selected 5-unit-cells edge-filter design has 93.6% transmittance in amorphous phase and 0.2% in crystalline phase, a 93.4% transmittance contrast with ~200 nm FWHM, with minimal optical degradation after enduring over 1.5×10⁶ phase switching cycles. We also show the tunability of this design by phase transition of each Sb₂Se₃ layer and increase the optical performance across 900-1200 nm range by optimizing each layer thickness.

1. Introduction

The term phase change material (PCM) can have different meanings depending on the research field. In energy storage, the term refers to material that store and release thermal energy through changes in its state of matter (latent heat), such as between solid and liquid states [1]. However, in electronics and photonics, the term phase change material refers to a solid-state material that changes its crystal phase when its temperature rises above transition temperature (T_{trans}). This change leads to a contrast of complex refractive index ($\tilde{n}$) between the initial and post-transitioned phases, where $\tilde{n}$ comprises of the real refractive index (n) and the imaginary extinction coefficient (k) parts [2]. The refractive index contrast between the two phases is an order of magnitude larger than contrast caused by applied electric field or light intensity (Kerr effect) [3], this property makes PCM ideal for photonic applications that require the modulation of optical performances. Examples include reconfigurable metasurface [2, 4-6], optical limiters [3, 7-11], optical switches [12, 13], optical filters [14-16], and absorbers [17, 18].

Phase change material can be classified by its behavior after phase transition. A volatile PCM will revert to its pre-transitioned phase if temperature falls below T_{trans} . But a non-volatile PCM will remain in its transitioned phase even if temperature falls below T_{trans} . Non-volatile PCM can be turned back to its pre-transitioned phase by melting it and then quenched at a speed faster than the molecules can organize into an orderly structure. Two examples of non-volatile PCMs are chalcogenides glasses called Ge₂Sb₂Te₅ (GST) and Ge₂Sb₂Se₄Te (GSST). GST is one of the earliest and most studied PCM [19]. It has large refractive index contrast between amorphous and crystalline phases, low T_{trans} when compared to other non-volatile PCM, T_{trans} ~145 °C to cubical phase, and T_{trans} ~200 °C to hexagonal phase [3, 20]. GST drawback is its large extinction coefficient in both visible and IR spectra which causes high optical loss from absorption. Meanwhile, GSST (Ge₂Sb₂Se₄Te₁), which is a derivative of GST that substitutes

some part of tellurium with selenium, also has large refractive index contrast but has a lower extinction coefficient which helps increase the optical efficiency. But GSST has a higher T_{trans} at ~ 242 °C [20]. An example of volatile PCM is vanadium dioxide (VO_2), which in the past 10 years has gathered attention as a volatile PCM with T_{trans} closest to room temperature at 68 °. It also has a relatively large refractive index contrast starting from near-IR (NIR), with small extinction coefficient (k) in the insulating phase (cold state), but its k became higher when transitioned to metallic phase (hot state) [21].

Compared to the other two non-volatile PCMs, antimony triselenide (Sb_2Se_3) is a relatively new material in the field of photonics, the initial research is in solar-energy applications. But recently, Sb_2Se_3 has gathered interest as a non-volatile PCM for photonic applications because of its near-zero absorption in the near-IR and IR spectra, high and stable refractive index contrast from visible to infrared spectra, and the transition temperature of Sb_2Se_3 is between GST and GSST at ~ 186 °C [22, 23]. Aside from these, Sb_2Se_3 also has a fast phase-switching speed in a microsecond range [19, 23, 24], and its phase-switching endurance is also high, with experimental demonstrations showing that it can reach as high as 10^6 to 10^7 cycles [23, 24]. These properties make Sb_2Se_3 an attractive material for low-loss, reconfigurable optical switch or optical circuit that can be used in optical waveguide [4-6, 13, 19].

Similar to optical limiter, switchable optical filter also operates in two-mode, where the intensity of incident light can be reduced depending on the current mode. In optical limiter case, the mode changing mechanism is self-activated, i.e., PCM needs to undergoes phase transition by the absorption from incident light [9, 10]. This is because the switching time from ON to OFF mode needs to be minimized so that as little high-intensity light can reach the eyes or detector as possible. But for optical switch, the switching time is not the main requirement and the switching mechanism can be from other mean such as joule heating [15], or light-induced heating using laser with wavelength outside of its intended operating bandwidth (e.g., if the device is intended to operates at IR wavelength, the phase transition can be activated by visible laser) [23]. Both optical filter and optical limiter required maximize transmittance and zero reflectance in ON mode and minimize transmittance in OFF mode for the best efficiency. But optical limiter further required absorptance in ON mode as a self-activating switching mechanism. The availability of these two switching mechanisms in switchable filter is beneficial because it eliminates the required absorptance in the operation bandwidth for inducing phase transition, which helps increase the efficiency of the design. The designs of these two type of devices can be a planar thin-film structure or a metasurfaces; planar structures are easier to design, optimize and fabricate, but metasurface structures offer more design flexibility [10]. For a planar designs, there are many methods to create high transmission and reduce reflection, such as: single-layer or multilayers quarter-wave antireflection designs, line-pass filters, or Fabry-Pérot resonant cavity [25].

In this work, we investigate Sb_2Se_3 as a material for a tunable optical filter in a planar thin-film designs. The intended operating wavelength is selected to be 1064 nm because it is from Nd:YAG laser which are commonly used in laboratory setups. At this wavelength, Sb_2Se_3 has zero absorption in amorphous phase and near-zero in crystalline phase. Its refractive index contrast is stable throughout the IR spectrum, which will allow for a broadband design. The goal is to create a low-loss, broadband, transmittance-tunable optical filter.

2. Theories

2.1 Optical properties of Sb_2Se_3

Antimony triselenide (Sb_2Se_3) is an orthorhombic chalcogenide non-volatile phase change material (PCM) characterized by quasi-one-dimensional ribbons. While its amorphous phase consists of a disordered network of distorted motifs, crystallization restores long-range order along the ribbon direction. This thermally driven transition can be triggered by nanosecond to microsecond optical or electrical pulses [19, 23], the process is reversible and have high endurance, with optical switching reaching 10^6 to 10^7 cycles [23, 24]. As an optical device,

Sb₂Se₃ enables repeatable toggling between high and low transmission modes, provided the local heating stays below the melting or ablation threshold [4, 24].

The complex refractive index ($\tilde{n}$) of any phase change material at wavelength λ is:

$$\tilde{n}_p(\lambda) = n_p(\lambda) + ik_p(\lambda) \quad (1)$$

where the real term n is the refractive index, the imaginary term k is extinction coefficient, and p denotes the phase which for Sb₂Se₃ can either be amorphous (*amo*) or crystalline (*cry*). Dispersive models, such as Tauc-Lorentz or multi-oscillator fits are often used to represent the measured n_p and k_p over the spectral range of interest. But in this work, a linear interpolation is used since all the dataset used have the measurement interval of less than 4 nm. Figure 1 shows their complex refractive indices from 300 to 1700 nm using dataset from Delaney, et al. [19], its complex refractive index at 1064 nm are: $\tilde{n}_{amo} = 3.370 + 0.000i$, and $\tilde{n}_{cry} = 4.308 + 0.010i$. The blue lines represent real refractive index (n), and red lines represent imaginary extinction coefficient (k). In amorphous phase (Fig. 1a), the refractive index increases from ~ 2.4 at 300 nm to a peak of ~ 4.0 at 600 nm and gradually decreases to a stable value of ~ 3.3 above 1200 nm, indicating a strong dispersion near the band edge of visible spectrum and a weaker dispersion in the NIR spectrum. The extinction coefficient starts from ~ 2.1 at 300 nm and dropped to zero at around 850 nm. In the crystalline phase (Fig. 1b), the refractive index at the UV wavelengths has around the same value as that of amorphous phase but reached a higher peak of ~ 5.1 at around 660 nm, it is then dropped to a stable of ~ 4.1 in the NIR spectrum. The k -index of crystalline phase peaked at around 3.1 in the near-UV spectrum and rapidly approached zero above 900 nm.

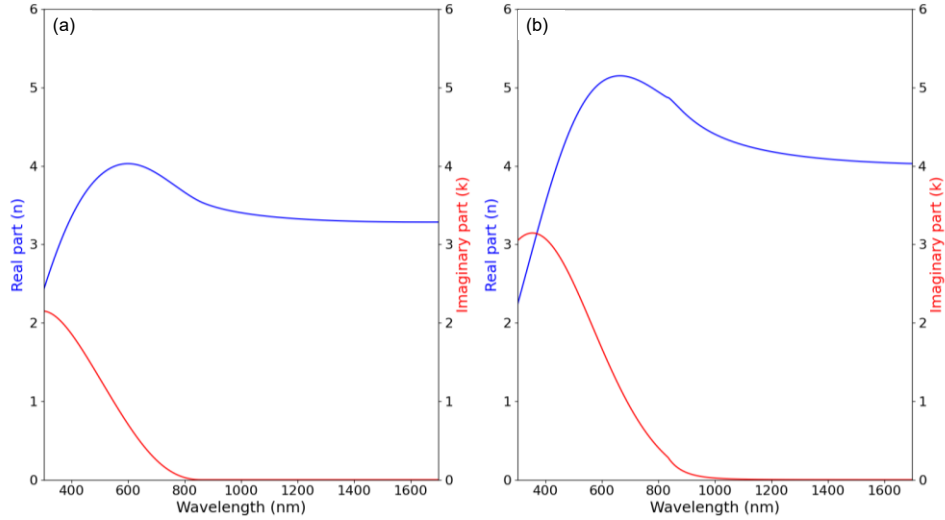


Fig. 1. Complex refractive index of Sb₂Se₃ in (a) amorphous, and (b) crystalline phases. Graphs are plotted from experimental data of Delaney, et al. [19].

The refractive index contrasts ($\Delta n = n_c - n_a$) are large and stable from the visible through IR spectra, and the extinction coefficients are large in the near-UV but falls to zero around NIR spectrum for both phases, with crystalline phase exhibiting higher absorption in the visible spectrum. So, when compared to other PCMs, Sb₂Se₃ has several advantages when operating in NIR and IR wavelengths. The large Δn and low k starting from NIR wavelengths enable a device to operate in two modes with little to no optical loss, stable Δn also helps with broadband designs.

2.2 Single-layer antireflection design

In designing most optical devices, the goal will be to either create antireflection (AR) or high reflection condition. For two-mode optical devices, e.g., tunable optical filters or optical limiters, both ON and OFF modes do not need reflection for any purpose, so the design goal will be to create antireflection. We can first consider the reflection in the ON mode and then consider the reflectance contrast (ΔR) between ON and OFF mode.

The simplest antireflection from a planar thin-film design is one film layer deposited on top of a substrate; both the film and substrate should have zero absorption. The diagram in the top right corner of Fig. 2 shows that a single-layer film reflects light at two interfaces; the bottom interface between film and substrate (R_1), and the top interface between air and film (R_2). Antireflection occurs from the destructive interference between R_1 and R_2 , with a complete AR condition requiring same amplitude and a phase difference of π radian (optical path length of $\lambda/2$). Since R_1 propagated through thin film layer twice (while transmitted down and reflected backup), the optical thickness (OT) of the film only needs to be equal to half of optical path length, $OT = \lambda/4$, also known as quarter-wave optical thickness (QWOT). Optical thickness is related to the physical thickness (t) by the following equation:

$$OT(\lambda) = n(\lambda)t, \quad (2)$$

where $n(\lambda)$ is the refractive index of the film at λ . For a single-layer AR device, the thin-film physical thickness (t) needs to be:

$$t = \frac{OT}{n} = \frac{\lambda/4}{n}. \quad (3)$$

Another method to calculate the thickness required for AR condition is to consider the reflection of the device. The reflectance (R) of just the substrate is related to its Fresnel coefficient (r) by:

$$R = |r|^2 = \left| \frac{n_a - n_s}{n_a + n_s} \right|^2, \quad (4)$$

where n_a is air refractive index, and n_s is substrate refractive index. If a single-layer film is placed on top of a substrate, the film and substrate layers can be combine to a so called effective reflectance index (n_r) [25], defined as:

$$n_r = n_f \frac{(n_s + n_f) + (n_s - n_f)e^{-i2\theta}}{(n_s + n_f) - (n_s - n_f)e^{-i2\theta}}, \quad (5)$$

where n_f is film refractive index, i denotes imaginary unit, and θ is phase thickness which is related to optical thickness (OT) and physical thickness (t) by angular wavenumber ($k_0 = 2\pi/\lambda$):

$$\theta = k_0 OT = k_0 nt. \quad (6)$$

By replacing n_s in Eq. 4 with n_r from Eq. 5, the reflectance of a single-layer device become:

$$R = |r|^2 = \left| \frac{n_a - n_r}{n_a + n_r} \right|^2. \quad (7)$$

Antireflection condition occurs at $R = 0$ or $n_r = n_a$. This condition happens when the phase thickness (θ) is in multiple of $\pi/2$ radian:

$$\theta = Z \frac{\pi}{2}, \quad (8)$$

where Z can be any even integer. If the condition in Eq. 8 is satisfied, Eq. 5 can be simplified with Euler's identity ($\exp^{-i\pi} = -1$) to:

$$n_r = n_f \frac{(n_s + n_f) - (n_s - n_f)}{(n_s + n_f) + (n_s - n_f)} = \frac{n_f^2}{n_s}. \quad (9)$$

By substituting n_r in Eq. 7 with Eq. 9, the required refractive indices for AR condition is:

$$n_a n_s = n_f^2. \quad (10)$$

Another condition is the relative refractive index between air, film, and substrate, which must be in the following order: $n_{air} < n_f < n_{substrate}$. Since light shifts its phase by π radian when reflected from lower to higher refractive index medium. The reflected light at air-film interface (R_2) will always undergo π radian phase shift since n_a will be the lowest in most cases. So R_1 must also undergo the same phase shift to create destructive interference.

These two conditions place a limitation on material selection since for a perfect antireflection, there is going to be only one ideal refractive index (n_{ideal}) for each n_s . In the case of Sb_2Se_3 , its refractive index is higher than substrate (silica; SiO_2) at every wavelength. So, if Sb_2Se_3 is QWOT thick, the reflection scenario has now become: R_1 still travelled 2QWOT (π radian) further than R_2 but only R_2 has its phase shifted during reflection by π radian, so the two R s have a phase difference of 0 radian or a constructive interference. To minimize reflection, Sb_2Se_3 optical thickness can be increased to two times the QWOT ($OT = \lambda/2$), this is called half-wave optical thickness (HWOT) or an absentee layer. As an example, at the reference wavelength $\lambda_0 = 1064$ nm, if we use 3-mm-thick SiO_2 as substrate ($n_s = 1.46$) [26], the ideal QWOT-film thickness is calculated from Eq.10 to be $n_{ideal} = 1.21$. A material with the closest refractive index is MgF_2 ($n_{MgF_2} = 1.38$) [27], a slight difference from ideal value causes the reflection to be higher, as illustrated in Fig. 2. The reflectance of HWOT-thick Sb_2Se_3 (blue) is the highest since it is just an absentee layer, the reflectance is equivalent to a standalone 3-mm SiO_2 substrate. An ideal film also exhibits some reflection because of the reflection at the substrate-air interface on the other side.

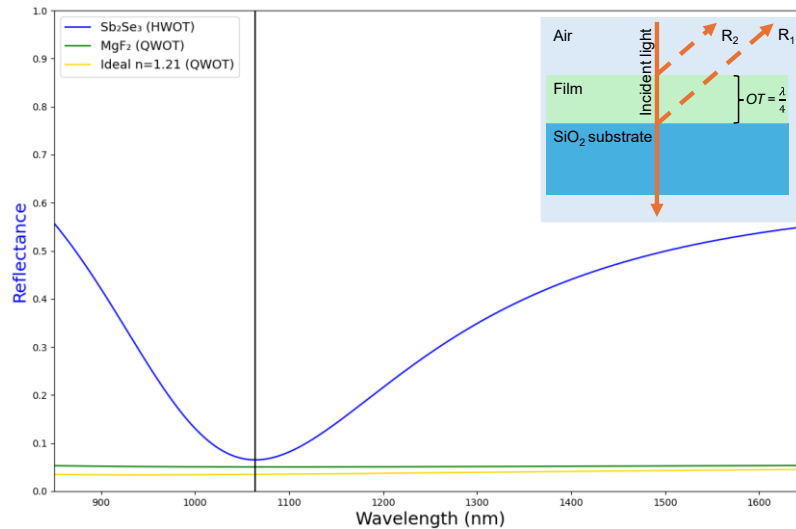


Fig. 2. Reflectance comparison of a single-layer design with different film. At 1064 nm (black, vertical line), HWOT-thick Sb_2Se_3 (blue) has the lowest reflectance $R = 6.5\%$. QWOT-thick MgF_2 (green) $R = 5.0\%$, slightly higher than an ideal film with $n_{ideal} = 1.21$ (grey) $R = 3.4\%$. The QWOT-thick structure is shown in the top right corner.

2.3 Multilayer antireflection design

A two-layer design with each layer thickness being QWOT (Q/Q design) can also be used to create antireflection condition. The Fresnel reflection coefficient (r) from Eq. 7 can be expanded to:

$$r = \frac{n_a - n_{r2}}{n_a + n_{r2}} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{f1}^2/n_s}\right)}, \quad (11)$$

where the numbering of n_f begins from substrate side and increases toward air side ($n_s / n_{f1} / n_{f2} / n_a$), n_{r2} is the effective reflectance index of the entire structure [25], and n_{r1} is the effective reflectance index of only substrate (n_s) and bottom film (n_{f1}). Similar to Eq.7, the antireflection condition of the two-layer design happens when $n_a = n_{r2}$ or:

$$n_a = \frac{n_{f2}^2}{n_{f1}^2/n_s}, \quad (12)$$

$$\frac{n_a}{n_s} = \frac{n_{f2}^2}{n_{f1}^2}. \quad (13)$$

From Eq. 13, it can be seen that the refractive index of the top film must be smaller than the bottom film ($n_{f2} < n_{f1}$). As an example, at 1064 nm if we use the same SiO₂ substrate ($n_s = 1.46$) and Sb₂Se₃ as bottom film ($n_{f1} = 3.37$), the ideal refractive index of the top film (n_{f2}) is going to be 2.78. A commonly used material with the closest refractive index is TiO₂ ($n_{f2} = 2.31$) [28]. When compared to single-layer Sb₂Se₃ design, two-layer Sb₂Se₃/TiO₂ design has lower reflectivity at 1064 nm, and broader AR region, as shown in Fig. 3. Another way to create two-layer AR condition is to fixed the two film refractive index and vary the thickness of each film instead, this is explored in chapter 5.2 of [25]. However, this method narrows the antireflection region.

Three-layer design with HWOT-thick film in the middle (Q/H/Q) required refractive index orderings are $n_s < n_{f1} < n_{f2}$ and $n_{f2} > n_{f3} > n_a$ for antireflection property. The middle layer (n_{f2}) acts an absentee layer when incident wavelength is the same as reference wavelength ($\lambda = \lambda_0$), but n_{f2} acts as a compensating layer to restore resonance in a scenario where λ differs greatly from λ_0 , a detailed explanation is available in chapter 5.4 of [25]. The reflection coefficient is equal to:

$$r = \frac{n_a - n_{r3}}{n_a + n_{r3}} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r2}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r2}}\right)}. \quad (14)$$

Because the middle film layer (n_{f2}) is HWOT thick, it behaves as an absentee layer [25, 29]. So $n_{r2} = n_{r1}$ and Eq. 14 becomes:

$$r = \frac{n_a - \left(\frac{n_{f3}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}, \quad (15)$$

a three-layer antireflection condition occurs when:

$$\frac{n_a}{n_s} = \frac{n_{f3}^2}{n_{f1}^2}. \quad (16)$$

The ratio of refractive index between top and bottom film is similar to a two-layer design in Eq. 13. As an example; using Eq. 16, if the bottom layer (n_{f1}) is Sb₂Se₃ (QWOT) then the ideal top layer ($n_{f3, ideal}$) is going to be 2.78, so TiO₂ ($n_{TiO2} = 2.31$) is used again. The middle layer needs to have the highest refractive index, so Silicon (Si) with $n_{Si} = n_{f2} = 3.55$ [30] is used. As shown in Fig.3, the three-layer design (yellow) reflectance at wavelength further from λ_0 (i.e., 1400 nm and longer) is lower than that of a single (blue) or two-layer design (green). This is the benefit of using Si as middle layer, it contribute to a broader operating bandwidth [29].

Instead of placing PCM as the bottom film layer, Sb₂Se₃ can be moved to the middle so that the selection of ideals n_{f1} and n_{f3} is easier. The bottom layer could be Si₃N₄ ($n_{f1} = 2.00$) [31] so

that the ideal top layer calculated from Eq. 16 have $n_{\beta} = 1.65$, which Al_2O_3 has the closest refractive index $n_{\text{Al}_2\text{O}_3} = 1.62$ [28]. These designs are shown in Fig. 3 where the reflectance of the single-layer, two-layer, and three-layer designs with Sb_2Se_3 placed directly on top of the substrate ($n_{\beta} = \text{Sb}_2\text{Se}_3$), are $\sim 6.5\%$. But the three-layer design with Sb_2Se_3 placed in the middle ($n_{\beta} = \text{Sb}_2\text{Se}_3$) has reflectance equal to 3.4%, the same value as an ideal single-layer design (3.4%) in Fig. 2. This improvement can be explained by the selection of n_{β} and n_{β} that are closer to ideal values.

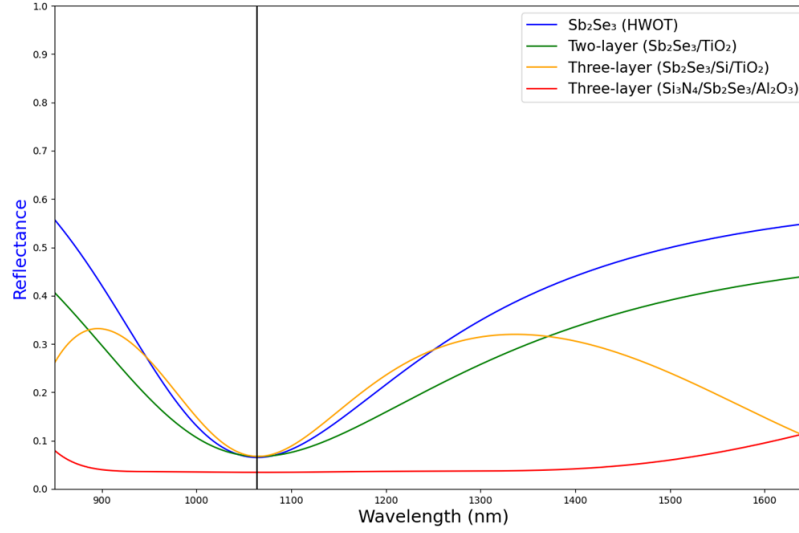


Fig. 3. Reflectance comparison between a single-layer and multi-layer Sb_2Se_3 designs. At 1064nm (black vertical line), the single-layer, two-layer and three-layer designs with Sb_2Se_3 as bottom layer (n_{β}) have similar reflectance at 6.5 - 6.7%. The three-layer design with Sb_2Se_3 as middle layer (n_{β}) has the lowest reflectance at 3.4%.

2.4 Long-pass edge-filter design

In all the previous examples, the reflectance after Sb_2Se_3 transitioned to crystalline phase have not been considered yet. Aside from low reflectance in ON mode, another requirement for transmittance tuning in optical device is high optical contrast between the two modes. Figure 4 shows the reflectance comparison between Sb_2Se_3 in amorphous (solid) and crystalline (dotted) phases, which represents the device in ON and OFF modes, respectively. These designs are the same as described in Fig. 3. The single-layer design has the highest reflectance contrast of 55.0% (Fig. 4a, blue). For the three-layer design with Sb_2Se_3 as the middle layer (Fig. 5b, red), even though it has the lowest reflectance in amorphous phase among other designs at 3.4%, its crystalline phase reflectance is only 14.8%, which make the reflectance contrast of only 11.4%.

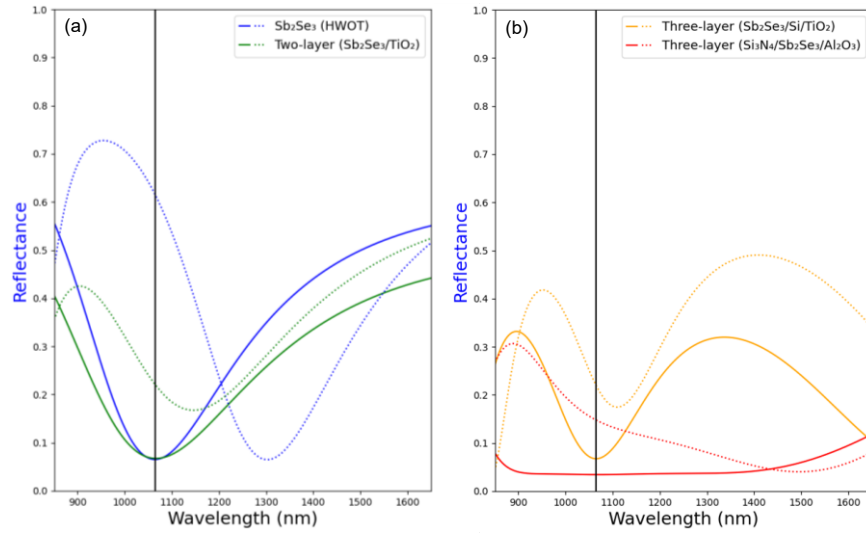


Fig. 4. Reflectance of the same designs shown in Fig. 3 before (solid) and after (dotted) phase transition. (a) Single-layer (blue) and two-layer design (green). (b) Three-layer designs with Sb_2Se_3 as the bottom layer (orange) and Sb_2Se_3 as middle layer (red). Single-layer design has the highest reflectance contrast. The vertical black lines indicate $\lambda_0 = 1064$ nm.

Another antireflection design is called an edge-filter design, it has high reflection in a region surrounding the reference wavelength (λ_0) but exhibits strong antireflection (AR) property outside of this region. The filter could be designed in such a way that the desired operating wavelength is in the AR region, at the edge of these two regions. When phase transition occurs and the device change to OFF mode, the reflectance profile will be shifted and the operating wavelength will be inside the high-reflection region instead. Edge filter design composed of a QWOT-thick high refractive index (H) and low refractive index (L) layers deposited symmetrically on top of each other in a structure called unit cell (U). One side of the AR regions will have a stable reflectance profile, determined by the structuring of the unit cell; a long-pass structure ($\frac{H}{2} L \frac{H}{2}$) has a stable AR condition wavelength longer than high-reflection region, and the short-pass structure ($\frac{L}{2} H \frac{L}{2}$) has a stable AR in a shorter wavelength, where $1/2$ denotes a layer with half-QWOT thickness. The number of unit cells could be one or more, with this number directly correlated to the steepness of regions' edges. To minimize the thickness of the device, a long-pass design is preferred since λ_0 will be at a shorter wavelength. And the spectral width of high-reflection region is directly correlated to the differences between H and L refractive index.

Figure 5 shows five long-pass edge filter designs with number of unit cells (N) ranging from 1 to 8. These designs use Sb_2Se_3 as H layer and MgF_2 as L layer, the reference wavelength is changed to $\lambda_0 = 740$ nm, and they are deposited on the same 3-mm-thick silica substrate. When these devices are in ON mode or amorphous phase (Fig. 5a) the area around the operating wavelength ($\lambda = 1064$ nm) is in antireflection region, but when devices switch to OFF mode or crystalline phase (Fig. 5b) the interested wavelength will be in high-reflection region. These shifts create a higher reflectance contrast values than any single or multi-layer designs, more than 90% contrast for 4, 6 and 8 unit-cell designs, much higher than any previously shown designs. It can also be seen that the increasing number of unit cells make the region edge steeper, but the improvement from 6 to 8 unit cells is lower than from 1 to 2 unit cells, so the benefits of increasing U should be weighted with the cons of increasing device thickness.

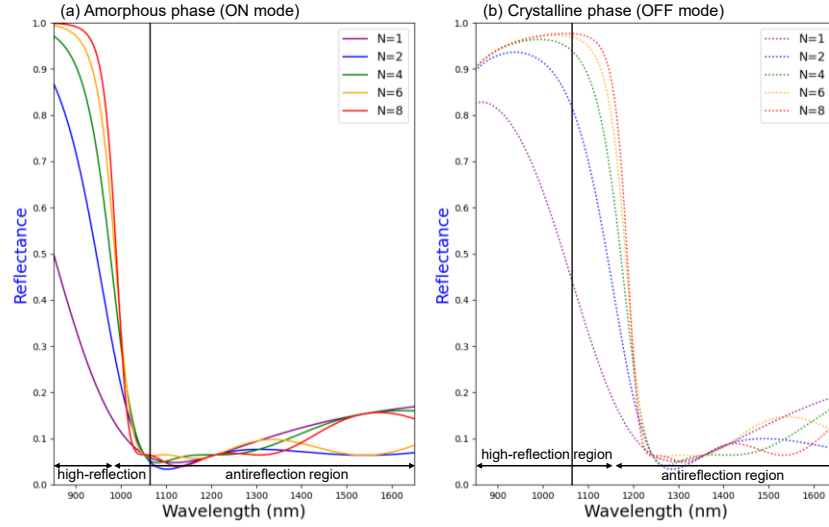


Fig. 5. Reflectance comparison of long-pass edge-filter designs ($\frac{H}{2}L\frac{H}{2}$) with varying number of unit cells. (a) When devices are in amorphous phase (ON mode), the interested wavelength (black vertical line) is in AR region. (b) When devices are in crystalline phase (OFF mode), the interested wavelength is in high-reflection region. Reference wavelength $\lambda_0 = 740$ nm.

3. Methods

3.1 Simulation of optical performances

In this work, we use transfer matrix method (TMM) of Fresnel equation to calculate the transmittance, absorptance and reflectance of every designs, the simulation code is based on [25], although a simulation software such as COMSOL, or OpenFilters [32] can also be used. We use 3-mm thick silica (SiO_2) as substrate ($n_s = 1.47$) [26], and also calculate the backside reflection from substrate-air layer on the other side. The high-index layers (H) are Si [33], TiO_2 [28], and Si_3N_4 [31], their refractive indices at 1064 nm are 3.56, 2.31, and 2.00, respectively. The low-index layers (L) are Al_2O_3 [28], SiO_2 and MgF_2 [27], and their refractive indices are 1.62, 1.45, and 1.38, respectively. All the above mentioned refractive index dataset can be found in *refractiveindex.info* website [34]. But because antimony triselenide (Sb_2Se_3) is a relatively new material in this field, its dataset cannot be found there. We utilize Sb_2Se_3 dataset from Delaney, et al. [19], (Fig. 1). The refractive index of air is set to a stable $n_a = 1.0$ across the spectrum. The incident angle is 0° and light is transverse electric (TE) polarized unless stated otherwise.

All of the designs are intended to operate at wavelength center around 1064 nm. The design comparison is between a single-layer, two-layers, three-layers, and edge-filter designs. The goal is to design a device with high transmission contrast (ΔT) between the two phases of Sb_2Se_3 by minimizing the reflectance in ON mode and maximize R in OFF mode.

3.2 Simulation of optical endurance and stability under large phase switching cycles

Experimental studies have demonstrated that Sb_2Se_3 can have phase switching cycles with minimal degradation to its optical contrast and maintain a low-loss, reversible modulation in an order of 10^6 cycles [23]. Even so, Sb_2Se_3 still exhibits slight complex refractive index drift due to its structural relaxation and cumulative thermal damage. Accounting for these variations is essential for ensuring long-term reliability and precision in high-endurance photonic applications [23, 24]. To account for this, the complex refractive index ($\tilde{n}$) can be rewritten from Eq. 1 to:

$$\tilde{n}_p(C) = n_p(C) + ik_p(C), \quad (17)$$

where C is the number of phase switching cycle.

The real part of the refractive index (n) is described by an exponential relaxation towards a saturated value such that:

$$n_p(C) = n_{p,\infty} - (n_{p,\infty} - n_{p,0})\exp^{-\alpha_p C}, \quad (18)$$

where $n_{p,\infty}$ is the asymptotic index after many cycles, $n_{p,0}$ is the refractive index at the first cycle (obtained from ellipsometry or literature data), and α_p is the drift coefficient per cycle of phase p [19, 35].

Meanwhile, the imaginary part (k) can be approximated from a linear growth as:

$$k_p(C) = k_{p,0}(1 + \beta_p C), \quad (19)$$

where $k_{p,0}$ is the initial loss and β_p is the loss growth coefficient. The parameter β_p is the relative loss per cycle of phase p [23, 36]. The model from Eq. (19) is adopted in this work, they are consistent with experimental trends and with general PCM endurance theory, providing a practical way to assess contrast degradation and threshold drift of PCM-based devices over large cycles [19, 23, 35, 36].

The optical endurance can be quantified using the following metrics: Transmitting contrast degradation (D), and insertion loss (IL). Transmittance contrast degradation is defined as the change in transmittance contrast after N cycles:

$$D(C) = \frac{\Delta T(C) - \Delta T_0}{\Delta T_0}, \quad (20)$$

where ΔT_0 is the transmittance contrast between amorphous and crystalline phases at the initial cycle $C = 0$ ($\Delta T_0 = T_a(0) - T_c(0)$). A negative D indicates a decrease in contrast.

The insertion loss (IL) of the device at cycle C compares the high transmittance amorphous phase after C cycle to an initial cycle:

$$IL_{dB}(C) = 10 \log \left[\frac{T_a(C)}{T_a(0)} \right]. \quad (21)$$

A growing $IL(C)$ indicates increasing loss, which is typically driven by the linear growth of the imaginary index $k_p(C)$ and by detuning of the resonance due to drift in $n_p(C)$.

4. Results & Discussions

4.1 Optical properties of 1064-nm optical filters

The design architectures, number of layers, total thickness of Sb_2Se_3 , the overall design thickness, transmittance in each mode, and transmittance contrast ($\Delta T = T_a - T_c$) of each design are shown in Table 1. The operating wavelength is 1064 nm, and the reference wavelength (λ_0) is 1064 nm for single-layer, two-layers, and three-layers designs, but λ_0 is changed to 740 nm for every edge-filter designs. Only single-layer design and edge-filter designs with at least two unit cells ($N \geq 2$) have a ΔT higher than 50%, with 7-unit-cells design having the highest $\Delta T = 95.1\%$. It is very important to note that higher unit cell numbers also require more Sb_2Se_3 to complete the resonance, a smaller Sb_2Se_3 has a benefit of reducing the energy required to switch the mode. And there is only marginal improvement in ΔT when N is increased to above 3, but the Sb_2Se_3 required and the overall device size increased drastically, e.g., ΔT improved by 2% in $N = 4$ design when compared to $N = 3$ design, the thickness increased by 33.3%, but when comparing $N = 5$ to $N = 4$ designs, ΔT increased by only 0.6% but the device needs to be 25.0% thicker.

In edge-filter designs, the transmittance in OFF mode decreases with increasing number of N (Fig. 5b), with $N \geq 6$ designs having zero transmittance. This is because N correlates to the total Sb_2Se_3 thickness and Sb_2Se_3 in crystalline phase has a small, but not zero extinction coefficient ($k_{\text{cry}}(1064) = 0.0097$), so, light is being absorbed by each Sb_2Se_3 layer. For reference,

this value is still an order of 10^2 smaller than other PCMs such as GST ($k_{GST, cry} = 2.4540$) or VO₂ ($k_{VO2, metallic} = 1.4580$).

Table 1. Designs of 1064-nm Sb₂Se₃-based optical filter. Transmittance is calculated at the interested $\lambda = 1064$ nm. Single, two, and three-layer designs all have $\lambda_0 = 1064$ nm, but all edge-filter designs have $\lambda_0 = 740$ nm.

Design	Layout	Number of layers	SbSe used (nm)	Device size (nm)	T (ON) (%)	T (OFF) (%)	ΔT (%)
Single-layer	SbSe	1	157.9	157.9	93.5	37.2	56.3
Two-layers	SbSe/TiO	2	78.9	194.1	93.3	76.9	16.4
Three-layers	SbSe/MgF/TiO	3	78.9	580.3	93.3	76.9	16.4
Three-layers	SiN/SbSe/AlO	3	157.9	455.7	96.6	83.3	13.3
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^1$	3	48.5	182.5	94.1	54.3	39.8
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^2$	5	97.0	365.0	95.0	16.3	78.7
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^3$	7	145.5	547.6	94.8	4.0	90.8
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^4$	9	194.0	730.1	93.7	0.9	92.8
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^5$	11	242.6	912.6	93.6	0.2	93.4
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^6$	13	291.1	1,095.1	94.5	0.0	94.5
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^7$	15	339.6	1,277.7	95.1	0.0	95.1
Edge filter	$(\frac{SbSe}{2}/MgF/\frac{SbSe}{2})^8$	17	388.1	1,460.2	94.4	0.0	94.4

Figure 6 compares the transmittance contrast (ΔT) between a single-layer design and edge-filter designs with $N \leq 6$ found in Table 1. Even though edge-filter designs with N from 3 to 6 (cyan, green, yellow, and orange, in increasing order) have similar transmittance contrast at the operating wavelength (black vertical line), but ΔT is higher and more stable with increasing number of N . This property should also be considered for broad spectrum operation.

Another two properties that might need to be considered are the optical performance at varying incident angle, the polarization mode of incident light (TE or TM). Figure 7 shows 4-unit-cells and 5-unit-cells designs with incident angle ranging from 0 to 90 degrees, and with polarization in TE (Fig. 7a and 7c) and TM (Fig. 7b and 7d). For both TE and TM polarized light, ΔT remains higher than 90% from 0° to ~30°, but TM polarization causes the high- ΔT region to blueshift greatly when increasing the incident angle. The spectral width of high- ΔT region in both designs are comparable, but 5-unit-cells design has a higher ΔT across the spectrum, which indicates a better broadband, wide-angle, tunable optical device.

With the reasons given above, we decided that 5-unit-cells, long-pass edge-filter design (Fig.6, yellow) has the best compromise between high optical performances, device size, and design complexity. This design has a broadband transmittance contrast between amorphous (ON mode) and crystalline phases (OFF mode), the FWHM of high- ΔT region is 197.0 nm. This design is going to be used as an example for the following sections.

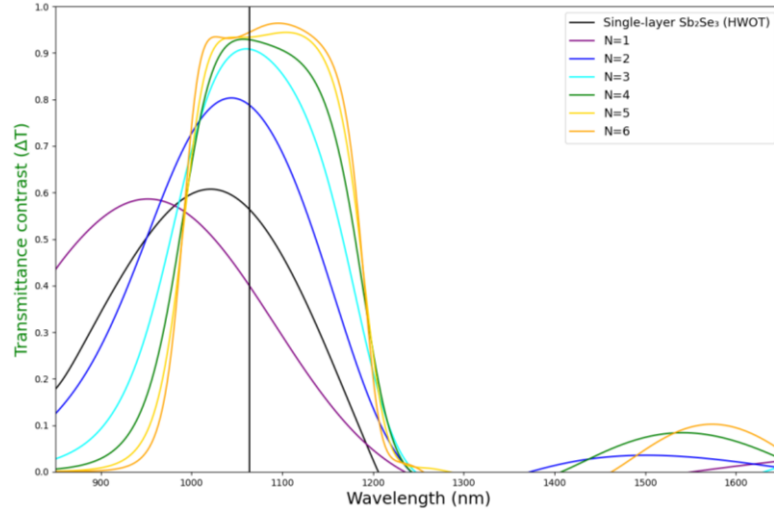


Fig. 6. Transmittance contrast (ΔT) of a single-layer and edge-filter designs with different number of unit cells (N). The vertical black line indicates the interested 1064 nm wavelength.

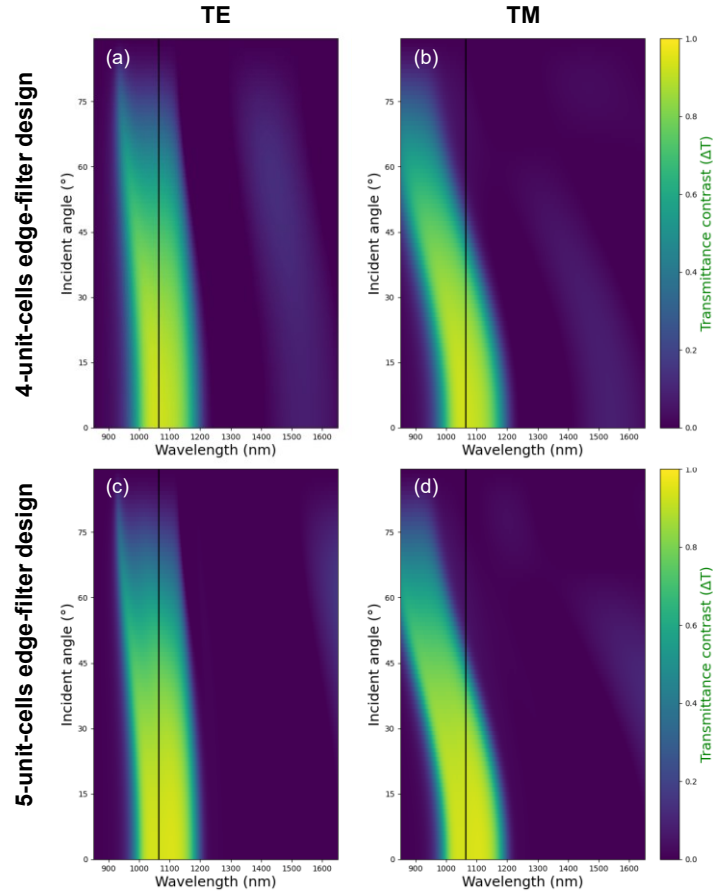


Fig. 7. Transmittance contrast at different incident angles and wavelengths of 4-unit-cells and 5-unit-cells edge-filter designs. The polarizations of incident light are TE (a, c) and TM (b, d).

4.2 Endurance simulation of 1064-nm optical filters(check endurance of edge-filter.)

Endurance testing of the single-layer HWOT-thick Sb_2Se_3 design, 4-unit-cells design and 5-unit-cells designs at 1064 nm demonstrated that all three resonant optical filters remain operational after 1.5×10^6 phase switching cycles, with design-dependent degradation of their transmission metrics. For the single-layer device, the amorphous phase transmittance (ON mode) stays close to its initial high value while the crystalline phase transmittance (OFF mode) changes modestly (Fig. 8a). Also, the corresponding percentage transmittance contrast degradation (Fig. 8d) and insertion loss (Fig. 8g) in amorphous phase increase slowly. These confirm that the single-layer design is robust against repeated switching with only minor deterioration of reducing amorphous phase transmittance. In the 4-unit-cells edge filter design, the amorphous phase transmittance also remains high but the crystalline transmittance exhibits slight increase (Fig. 8b). Which cause the contrast degradation, shown in Fig. 8e, to has a noticeably larger change than the single-layer design, which reflects the higher sensitivity of resonant edge to small changes in Sb_2Se_3 refractive index. The insertion loss of amorphous phase (Fig. 8h) also increase more rapidly, which consists of a **stronger field confinement and longer effective optical path within the lossy material**. The 5-unit-cells structure further accentuates these trends: its crystalline-phase transmittance (Fig. 8c) and transmittance contrast degradation (Fig. 8f) have the largest change among the three designs, and its insertion loss (Fig. 8i) also shows the largest cumulative increase.

These results highlight the expected trade-off between high initial optical performance and endurance robustness: increasing cavity complexity and Q-factor enhances on-resonance modulation but also amplifies the impact of endurance-induced index drift and loss in Sb_2Se_3 . A simpler single-layer stack provide better long-term stability, whereas resonant designs require tighter control of material and thermal cycling conditions. Even so, both long-pass edge-filter designs still has a higher ΔT after 1.5×10^6 cycles than the single-layer design.

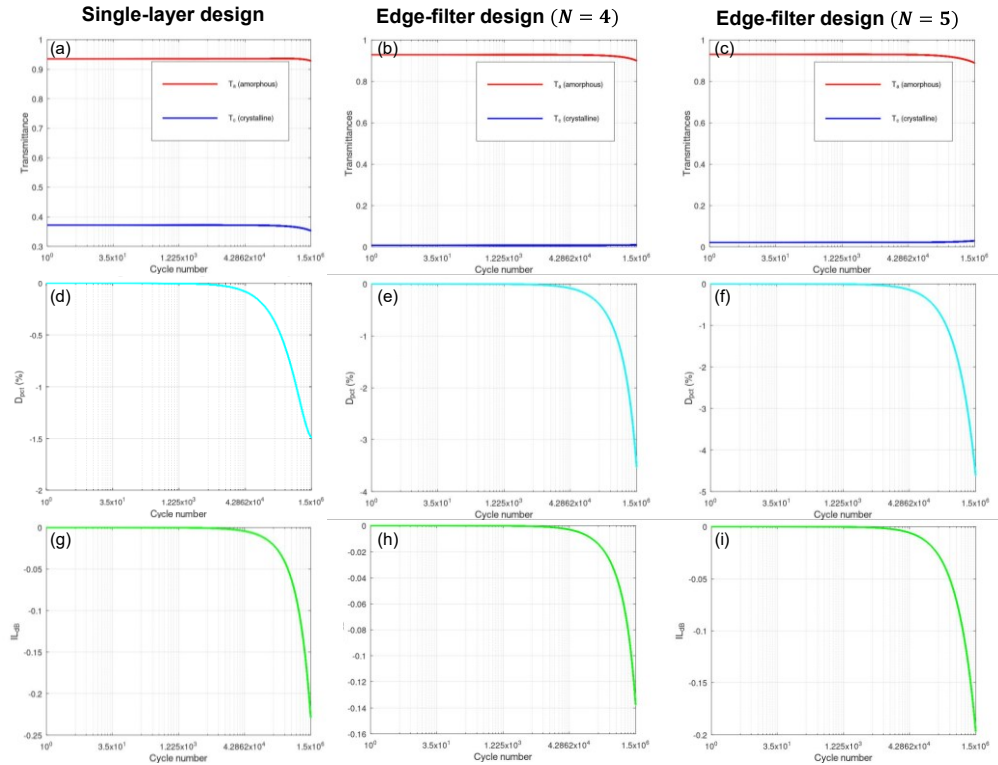


Fig. 8. Endurance simulation of tunable 1064-nm optical filter after 1.5×10^6 phase switching cycles. (a-c) The transmittance in ON mode (red) and OFF mode (blue) phases, (d-f) the transmittance contrast degradation, and (g-i) the ON mode insertion losses of each design. Note the different y-axis scales.

4.3 Transmittance tunability and optimization.

Edge-filter designs have multiple layers of Sb_2Se_3 which complicates the phase switching setup. When laser induced heating is used as a switching method, the phase transition of every Sb_2Se_3 layer would not occur simultaneously, and the transmission will change in a gradual manner. This effect is shown in Sarangan, et al. [3]'s optical limiter, their device has a similar long-pass edge-filter design $(\frac{H}{2}L\frac{H}{2})^4$ but H layers are GST instead of Sb_2Se_3 , and L layers are SiO_2 instead of MgF_2 . Their experimental result shows that the phase switching of GST layers happens layer-by-layer in an order that is determined by the changes in optical field distribution and thermal crosstalk between layers, which does not necessarily start from air-side to substrate-side. However, our device is intended to be used as a transmittance tunable filter which does not require a fast switching speed, so electrical heating is also available as an alternative phase switching method. The ability to manually control the phase of each Sb_2Se_3 layer will enable a filter to have multiple transmittance value. Using the 5-unit-cells edge-filter design described in Table 1, which consists of 11 layers of Sb_2Se_3 and MgF_2 (6 layers being Sb_2Se_3), there would be a total of 2^6 configurations for Sb_2Se_3 phases. Figure 9 highlights some of these configurations, the legend denotes Sb_2Se_3 in amorphous (a) or crystalline (c) phase and the ordering from left to right is substrate to air side (f_1 to f_6). With the correct phase configuration, the transmittance can be gradually reduced from peak 93.4% to 0.0%.

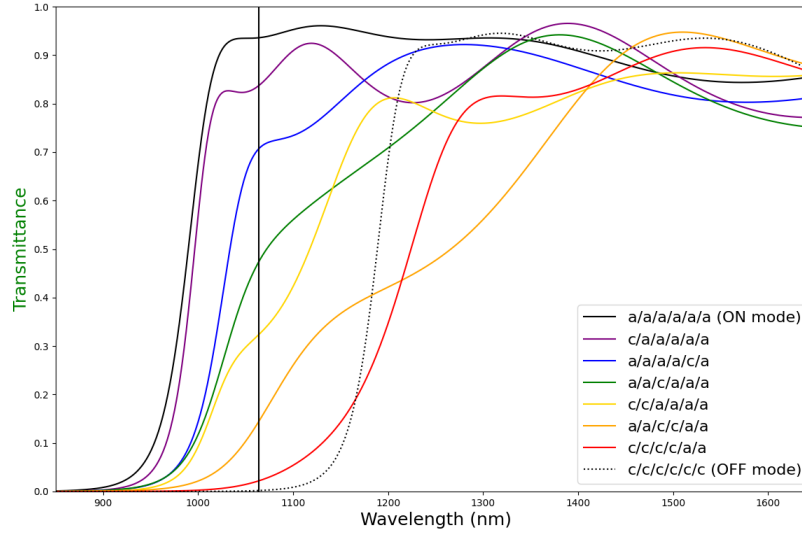


Fig. 9. Transmittance of the five-unit-cells edge-filter design in 8 different configurations. Each Sb_2Se_3 layer is in amorphous (a) or crystalline (c) phases, and the layout ordering from left to right is substrate-side to air-side.

The peak transmittance contrast of each design shown in Fig. 6 does not always be at the desired operating wavelength of 1064 nm. This is explained by the arbitrary selection of reference wavelength in edge-filter designs. An optimization algorithm can be used to improve the optical performance at the desired wavelength, a simple merit function found in OpenFilters

[32] is used as an example here. The optimization goal for the 5-unit-cells edge-filter design is to have a transmittance value of at least 97% from 1000 to 1200 nm in an interval of every 20 nm. The optimization took less than 5 seconds and the new design now required 304.2 nm of Sb_2Se_3 , 25% more than the original design, but the overall thickness is now 845.6 nm, 93% of the original (the actual thickness of each layer is shown in Table 2). Figure 10a shows the transmittance of the original (green) and optimized design (blue) in amorphous (solid) and crystalline phase (dotted). Overall, the optimized design has higher transmittance at the desired 1000 to 1200 nm spectrum, but the spectral profile has been blueshifted by a small amount. The transmittance contrast (Fig. 10b) of the optimized design is also higher at 1064 nm, 95.8% compared to 93.4% of the original. The transmittance contrast FWHM has also slightly widen to 207.4 nm (original: 197.0 nm).

Table 2. Thickness of each layer before and after optimization. Sb_2Se_3 and MgF_2 layers units are in nm.

Design	SiO_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3	MgF_2	Sb_2Se_3
Original	3 mm	24.3	134.0	48.5	134.0	48.5	134.0	48.5	134.0	48.5	134.0	24.3
Optimized	3 mm	46.1	67.5	59.3	148.1	55.0	89.3	50.0	145.3	63.3	91.2	30.5

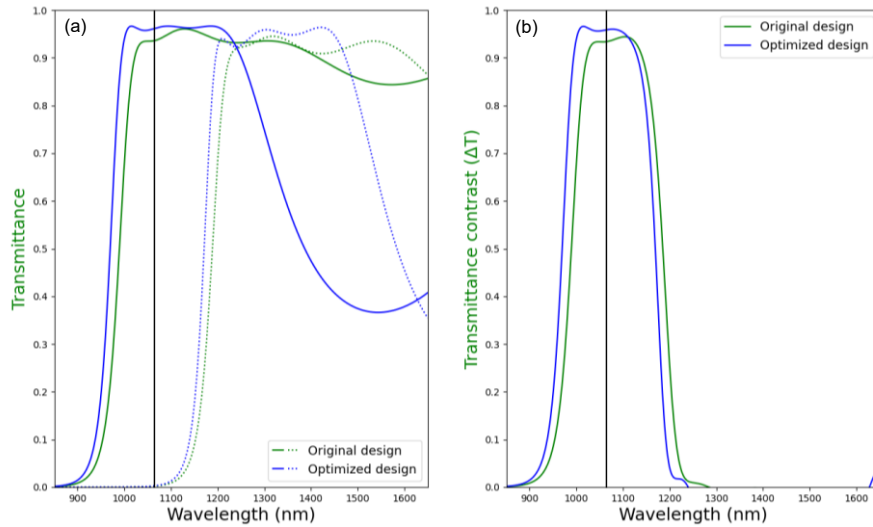


Fig. 10. Transmittance comparison of 5-unit-cell edge-filter design before (green) and after (blue) optimization. (a) The transmittance in ON mode (solid) and OFF mode (dotted). (b) The transmittance contrast (ΔT). An optimized design has higher optical performance from 900 to 1200 nm, but the high- ΔT region has been blueshifted by a small amount.

5. Conclusions

We have shown that Sb_2Se_3 is a suitable PCMs for a high-efficient, broadband, transmittance tunable, optical filter operating around 1064 nm wavelengths. In our opinion, the long-pass edge-filter with 5-unit-cells has the best compromise between optical performance, size, and design complexity. This design required 6 layers of Sb_2Se_3 and 5 layers of MgF_2 and is overall 912.6 nm thick. The transmittance at 1064-nm in ON mode is 93.4% and 0.2% in OFF mode ($\Delta T = 93.4\%$), the FWHM of ΔT is around 200 nm. The transmittance contrast degradation and insertion loss after simulating 1.5×10^6 phase changing cycles are within acceptable ranges. The multilayer nature of edge-filter design enables the transmittance value to be modulated

gradually, and using optimization algorithm in OpenFilters, the optical performances from 900 to 1200 have been increased.

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Data availability. Data underlying the results presented in this paper may be obtained from the corresponding author upon a reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

The authors used Gemini to check the logic and grammar of the manuscript, and Consensus was used for the initial literature review. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

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